

Zacharie Garnier-Cuchet

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 [zachariegarnier](#) |  [zachariegarniercuchet](#) |  Zacharie Garnier-Cuchet

Montréal, Québec - H2H2G8, Canada

EXPERIENCE

• Polytechnique Montréal

2025 – Present

NLP Teaching Assistant

Montréal, Canada

- Assisting in the Master's-level Natural Language Processing course, supporting students in implementing neural models and foundational NLP algorithms.
- Designed and corrected assignments involving Word2Vec, SGNS, GPT and PyTorch implementations.
- Guided student projects and provided technical feedback on model performance and reproducibility.

• Université de Technologie de Compiègne (UTC)

2022 – 2024

Mathematics Teaching Assistant

Compiègne, France

- Taught calculus, linear algebra, and probability to students with disabilities, adapting methods to diverse learning needs.
- Provided personalized academic support, improving student success and autonomy.

• Bombardier Produits Récréatifs (BRP)

Sep 2023 – Feb 2024

Industrial Engineering Intern (New Product Development)

Valcourt, Québec, Canada

- Automated industrial processes for Can-Am vehicle development by streamlining project deliverables and data workflows.
- Developed tools improving data traceability and reducing manual processing time.
- Contributed to new product development pipelines in a fast-paced engineering environment.

• IBM Quantum

Feb 2023 – 2023

Quantum Computing Intern

Remote

- Conducted introductory research in quantum computation under IBM Quantum mentorship.
- Explored quantum circuits and algorithms, gaining hands-on experience with cloud-based quantum hardware.

• Focal-JMlab

Jan 2021 – Feb 2021

Manufacturing and Industrial Operations Intern

La Talaudière, France

- Worked in production workshops and logistics warehouses, learning practical industrial processes.
- Assisted with supply chain workflows and quality control tasks.

EDUCATION

• Polytechnique Montréal / MILA – Québec AI Institute

Sept. 2024 – July 2026

M.Sc.A. (Research) in Computer Engineering – Artificial Intelligence

Montréal, Canada

- GPA: 4.0/4.0
- Coursework: Natural Language Processing, Deep Learning Dynamics, Reinforcement Learning, Representation Learning, Stochastic Processes, Data Science.
- Research focus: Theoretical and applied NLP.

• Université de Technologie de Compiègne (UTC) / Sorbonne University Alliance

Sept. 2020 – June 2024

Engineering Degree (Master-level) in Mathematics, Physics, and Mechanical Engineering

Compiègne, France

- GPA: 5.0/5.0
- Coursework: Thermodynamics, Fluid Mechanics, Mechanical Vibrations, Structural Optimization.

• Technische Universität Berlin (TUB) / Berlin University Alliance

Mar. 2022 – Aug. 2022

Academic Exchange Semester – Mechanical Engineering

Berlin, Germany

- Coursework: Linear Algebra, Probability Statistics, Applied Physics, Mechanical Engineering fundamentals.
- Completed an intensive German language program and explored international research perspectives.

PROJECTS

- **Applying RLHF to Humor Generation - Reinforcement Learning Course Project** Sept. 2024 – Dec. 2024
Tools: PyTorch, TRL, PPO, Phi-3-Instruct, BERT Reward Model 
 - Implemented a full RLHF pipeline to align a language model toward humorous dialogue generation.
 - Trained a BERT-based reward model to evaluate humor quality and guide policy optimization.
 - Applied Proximal Policy Optimization (PPO) via the TRL library to fine-tune Microsoft Phi-3.
- **Dynamic Analysis of Taipei 101 - Structural and Modal Vibrations Study** Mar. 2024 – Aug. 2024
Tools: MATLAB, Python, Finite Element Methods 
 - Modeled the Taipei 101 tower using Rayleigh–Ritz formulations and finite element abstractions.
 - Computed eigenmodes, modal frequencies, and compared them with theoretical expectations.
 - Analyzed the efficiency of passive mass dampers and their impact on structural response.
 - Developed numerical simulations to interpret complex dynamic behaviors under external excitation.
- **N-SAT Solver - Independent Algorithmic Project** Summer 2022
Tools: Python, Backtracking, Heuristics
 - Designed and implemented a solver for generalized N-SAT problems with custom heuristics.
 - Analyzed worst-case and empirical time complexity across different constraint structures.
 - Applied the solver to real combinatorial tasks (Sudoku, Nonograms) with successful results.
 - Explored pruning strategies and partial assignments to reduce exponential branching.
- **Optimal Wordle Solver - Independent Algorithmic Project** Mar. 2022 – Jun. 2022
Tools: Python, Search Optimization, Information Theory
 - Built a Wordle-solving algorithm based on entropy reduction and constraint filtering.
 - Reduced runtime from hours to seconds through optimized data structures and pruning.
 - Analyzed quadratic complexity and implemented improvements for near real-time inference.
- **Text Decoder via Stochastic Key Search - Independent Project** 2021
Tools: Python, Randomized Algorithms, Heuristic Search
 - Developed a decoder for a substitution-based encoding system assigning unique integers to characters.
 - Designed a stochastic exploration algorithm to navigate the space of possible keys.
 - Implemented heuristic minimization criteria to converge on coherent plaintext solutions.
- **Nim Game with Intelligent Agent - Independent Project** Jan. 2024 – Feb. 2024
Tools: Google Sheets, Combinatorial Game Theory
 - Implemented an intelligent adversary for the Nim game inside Google Sheets.
 - Used winning-position theory (XOR heap invariants) to guarantee optimal play.
 - Logged game histories for post-hoc analysis of strategy effectiveness.
- **Exploration of the Syracuse (Collatz) Conjecture - Independent Math Project** 2020
Tools: Python, Discrete Dynamical Systems
 - Investigated structural properties of the Syracuse sequence and constructed analyzable variants.
 - Explored convergence patterns and produced a structured mathematical report supervised by a physicist.
 - Applied numerical experimentation to study density and stopping-time behaviors.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION

- [C.1] Z. Garnier-Cuchet, et al. (2025). **Revisiting the Goldilocks Zone in Inhomogeneous Networks**. In ICML 2025 Workshop on High-Dimensional Learning Dynamics (HiLD). Jun. 2025.

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** PyTorch, JAX, HuggingFace Transformers, Lightning, Weights&Biases, Optuna; Contrastive learning, metric learning, retrieval models, reinforcement learning, sequence modeling, representation learning.
- **NLP & Information Retrieval:** Tokenization (BPE, SentencePiece), SGNS/Word2Vec, GloVe, BM25, ColBERT, LLM-based encoders, cross-encoders, rerankers, long-context models, legal NLP, citation retrieval, sequence-to-sequence models.
- **Scientific Programming & Data:** Python, C, R; NumPy, Pandas, Scikit-learn; data processing at scale, vector stores, approximate nearest neighbors (FAISS).
- **Mathematics & Statistics:** Probability theory, stochastic processes, Monte-Carlo methods, numerical optimization, convex analysis, high-dimensional geometry, linear algebra, statistical modeling.
- **Software & Engineering:** Linux, Git, Docker, SLURM/HPC, Conda, Make; API design, profiling, testing; version-controlled research pipelines.
- **Cloud & Deployment:** AWS (EC2, S3), GCP, Compute Canada, Kubernetes (basic), REST backends.
- **Research Skills:** Experimental design, ablation studies, reproduction of baselines, dataset construction, annotation guidelines, metric analysis, error analysis, LaTeX for scientific writing.

LEADERSHIP & PROFESSIONAL EXPERIENCES

- **Leadership & Service**
 - **President, BRP Intern Committee (2023)** - Represented 100+ interns; coordinated events and communication with upper management.
 - **Class Representative (France, 3 years)** - Academic mediation, conflict resolution, and cohort coordination.
 - **Activities Lead, Student Association (UTC)** - Managed events, logistics, and student engagement.
- **Professional Experience**
 - **Cook & Logistics Manager, Pizzeria (Ireland)** - Paid position following Workaway; managed kitchen operations, supply chain, and service (june-sept 2023).
 - **Café Service & Operations Assistant, Local Café (Montréal)** - Customer service and daily operations (Jan-Mar 2023).

ADDITIONAL INFORMATION

- **Languages:**
 - French (native)
 - English - daily use; learned at school and during stay in Ireland
 - German - multiple travels; 10+ years learning at school
 - Spanish - travel in South America; self-taught and one formal course
- **Sports:**
 - Marathon: Paris 2023 (3:05), Petit Train du Nord 2024 (1st in category), Montréal 2025 (pacer)
 - Half marathons: Montréal 2023-2024
 - Trail: Mont-Tremblant 2025 (14/300)
 - Other: Tennis (15/3), climbing, handball, soccer
- **Music:**
 - Trombone (11 years), Piano (5 years), Jazz improvisation, Composition
 - Member of jazz Big Band and various ensembles
 - Former conservatory student; performer in a symphonic orchestra